MDS-005 — Motion System

External publication draft

MDS-005 — Motion System

> *Motion is not animation. Motion is the visible expression of understanding changing over time.*

Purpose

The previous Mosaic Design System specifications established:

- Design Tokens
- Colour
- Materials
- Typography

These define what the interface is.

MDS-005 defines **how that interface changes**.

Unlike conventional motion systems, which primarily focus on visual transitions, the Mosaic Motion System exists to communicate behavioural continuity.

Motion should explain:

- what changed,
- why it changed,
- how it relates to the previous state.

The user should never perceive animation.

They should perceive understanding.

Relationship to Previous Specifications

```text
Vision

↓

Principles

↓

Mental Model

↓

Interaction

↓

Composition

↓

Tokens

↓

Colour

↓

Materials

↓

Typography

↓

Motion

↓

Components

↓

Presentation
```

The Motion System consumes:

- Interaction Model
- Composition
- Material System
- Typography

It communicates behavioural evolution over time.

Scope

This specification defines:

- Motion Philosophy

- Motion Hierarchy
- Behavioural Motion
- Material Motion
- Refraction Motion
- Temporal Continuity
- Motion Curves
- Motion Accessibility
- Runtime Motion Resolution
- Platform Motion

This specification intentionally does **not** define:

- Components
- Layout
- Business Logic
- Interaction Behaviour

Those systems drive motion.

They do not define it.

Core Question

MDS-005 exists to answer one question.

> ***How should behavioural change become visible?***

Not:

> *Which animation should play?*

Motion Statement

Within Mosaic:

> ***Motion communicates understanding.***

If removing motion reduces understanding...

The motion belongs.

If removing motion changes only appearance...

The motion should be questioned.

Motion Responsibilities

The Motion System separates motion into several conceptual layers.

```
[text]
Behaviour

↓

Temporal Meaning

↓

Material Response

↓

Motion

↓

Presentation
```

Each layer contributes one responsibility.

Motion is therefore the consequence of behaviour rather than an isolated visual system.

Expected Outcome

After reading MDS-005 contributors should understand:

- why Mosaic moves,
- how behaviour drives motion,
- how materials participate,
- how continuity is preserved,
- how accessibility affects motion,
- how runtime motion is resolved,

without discussing rendering technologies or animation frameworks.

Repository Structure

```
[text]
design/

■■■ mds/

    ■■■ MDS-005 Motion System/

        README.md

        00-document-control.md

        01-motion-philosophy.md

        02-motion-hierarchy.md

        03-behavioural-motion.md

        04-material-motion.md

        05-refraction-motion.md

        06-temporal-continuity.md

        07-motion-curves.md

        08-accessibility.md

        09-runtime-motion-resolution.md

        10-platform-motion.md

        11-governance.md

        12-adrs.md

        13-contributor-guidance.md

        glossary.md

        references.md
```

Dependencies

Required reading:

- MDL-001 → MDL-005
- MDS-001 Design Token Architecture
- MDS-002 Colour System
- MDS-003 Material System
- MDS-004 Typography System

Downstream specifications:

- MDS-006 Composition Engine
- MDS-007 Tile Framework
- MDS-008 Component Library

Review Status

Status

Draft

Owner

Lead Design Systems Architect

Next File

`00-document-control.md`

Document Control

Document Information

Property	Value
Document ID	MDS-005
Title	Mosaic Design System — Motion System
Classification	Internal
Status	Draft
Version	0.1
Owner	Lead Design Systems Architect
Parent Specifications	MDL-001 → MDL-005, MDS-001 → MDS-004
Repository	`/design/mds/MDS-005 Motion System/`

Purpose

MDS-005 defines the Motion System used throughout Mosaic.

Motion is not treated as visual decoration.

It is treated as the visible expression of behavioural change.

Where the Interaction Model explains:

> ***Why the user's World changes.***

The Motion System explains:

> ***How that change should become perceptible.***

Its responsibility is to transform behavioural evolution into physical continuity.

Authority

MDS-005 governs:

- Motion philosophy
- Behavioural transitions
- Material motion
- Refraction movement
- Temporal continuity
- Motion hierarchy
- Runtime motion resolution
- Motion accessibility

- Platform motion

This specification intentionally does **not** govern:

- Interaction behaviour
- Business logic
- Component implementation
- Layout

Those systems initiate change.

Motion communicates it.

Relationship To MDS

Motion sits after Materials and Typography because both participate in movement.

```
[mermaid]
flowchart TD

Vision
Vision --> Principles
Principles --> MentalModel
MentalModel --> Interaction
Interaction --> Composition
Composition --> Tokens
Tokens --> Colour
Colour --> Materials
Materials --> Typography
Typography --> Motion
Motion --> Components
Components --> Presentation
```

Motion consumes:

- Behaviour
- Composition
- Materials
- Typography

It communicates:

- continuity
- hierarchy
- physical evolution

Design Intent

Many motion systems begin by asking:

> *Which animation should play?*

Mosaic intentionally asks:

> *What changed?*

Motion therefore begins with behaviour rather than animation.

Animation is merely one possible implementation.

Understanding remains the architectural objective.

Reader Expectations

Before reading this specification contributors should already understand:

- MDL-001 Vision
- MDL-002 Principles
- MDL-003 Mental Model
- MDL-004 Interaction Model
- MDL-005 Composition Model
- MDS-001 Design Token Architecture
- MDS-002 Colour System
- MDS-003 Material System
- MDS-004 Typography System

The Motion System assumes those concepts already exist.

Its responsibility is to make them perceptible over time.

Architectural Scope

The Motion System defines:

- behavioural motion
- material response
- temporal evolution
- runtime motion
- accessibility
- platform adaptation

It intentionally avoids implementation-specific concerns such as:

- CSS animations
- Flutter animation controllers
- SwiftUI transitions
- Compose animation APIs
- easing function syntax

These are implementation artefacts.

The Motion System defines only the architectural model.

Stability

Expected lifetime.

Artefact	Expected Lifetime
Animation APIs	Months
Motion Curves	Years
Motion Hierarchy	Years
Behavioural Motion	Decades
Motion Philosophy	Decades

Rendering technology may evolve.

Motion behaviour should remain recognisably Mosaic.

Success Criteria

MDS-005 succeeds when:

- users always understand what changed
- movement preserves continuity
- materials move believably
- typography remains readable during transitions
- accessibility remains uncompromised
- motion quietly reinforces understanding rather than demanding attention

Users should rarely notice animation.

They should simply feel that the interface naturally evolved.

Review Status

Status

Draft

Dependencies

- MDL-001 → MDL-005
- MDS-001 → MDS-004

Supersedes

None.

Next File

`01-motion-philosophy.md`

Motion Philosophy

Purpose

Before defining curves, transitions or runtime behaviour, contributors must first understand what Motion represents within Mosaic.

Most interface systems treat motion as animation.

Animation is viewed as:

- delight,
- feedback,
- polish,
- visual quality.

Mosaic intentionally rejects this perspective.

Motion exists to communicate understanding.

It is part of the behavioural language of the platform.

Without Motion, users should still understand the interface.

With Motion, that understanding should become effortless.

Philosophy Statement

> ***Motion should explain behavioural change while preserving continuity, never drawing attention to itself.***

Everything within the Motion System derives from this statement.

Motion Is Communication

Motion communicates:

- change,
- continuity,
- hierarchy,
- causality,
- physicality.

It should never exist purely because movement appears visually attractive.

If removing Motion reduces understanding...

The Motion belongs.

If removing Motion only changes aesthetics...

The Motion should be questioned.

Motion Begins With Behaviour

Motion should never originate from interface events.

Incorrect.

```text
Button Click

↓

Animation
```

Preferred.

```text
Behaviour Changes

↓

Motion Explains Behaviour

↓

Presentation
```

Behaviour always possesses higher authority than animation.

This principle is inherited directly from the Interaction Model.

Motion Is Continuity

The user's World should feel continuous.

Motion preserves that continuity.

Instead of:

```text
Old State

↓

Replace

↓

New State
```

Mosaic prefers:

```text
Old State

↓

Evolve

↓

Understanding

↓
```

`New State`

Users should perceive evolution.

Not replacement.

Motion Is Physical

Because Mosaic possesses a Material System...

Motion should feel physical.

Examples include:

- Acrylic carrying momentum,
- Refraction redistributing gradually,
- Overlay Materials emerging naturally,
- Hero Materials settling into place.

Nothing should feel disconnected from the physical world established by the Material System.

Motion Supports Composition

Motion should reinforce the Composition Model.

Examples.

Hero changes.

↓

Hero moves first.

Supporting information follows.

Peripheral information settles last.

Movement reinforces editorial hierarchy.

It never competes with it.

Motion Supports Reading

Typography should remain readable throughout movement.

Poor.

Fast transitions.

↓

Text becomes difficult to read.

Preferred.

Typography remains stable.

↓

Movement occurs around it.

The interface should continue feeling editorial even while changing.

Motion Supports Atmosphere

Runtime Atmosphere should evolve through motion.

Example.

Artwork changes.

↓

Atmosphere blends.

↓

Materials respond.

↓

Refraction redistributes.

↓

The environment settles.

Users should perceive environmental evolution rather than colour replacement.

Motion Is Restrained

The Motion System intentionally values restraint.

Avoid:

- exaggerated overshoot,
- playful bounce,
- unnecessary elasticity,
- decorative choreography.

The interface should feel:

- confident,
- deliberate,
- physically believable.

Entertainment already provides excitement.

Motion should provide clarity.

Motion Is Predictable

Users should gradually learn how Mosaic moves.

After sufficient use they should instinctively predict:

- where objects come from,
- where they go,
- what changed,
- why it changed.

Predictability reduces cognitive effort.

Novelty should never become a design objective.

Motion Is Layered

Motion should occur at multiple conceptual layers.

```
[text]
Behaviour

↓

Composition

↓

Materials

↓

Typography

↓

Presentation
```

Every layer contributes movement.

No layer should dominate the others.

Motion Has Meaning

Every movement should answer at least one question.

Examples.

Why did this appear?

↓

It became relevant.

Why did this move?

↓

Hierarchy changed.

Why did this disappear?

↓

Its purpose ended.

If contributors cannot answer these questions...

The movement probably should not exist.

Motion And Time

Motion should communicate temporal progression.

Examples.

Playback.

↓

Episode complete.

↓

Next episode.

↓

Continuation.

The movement should make the passage of time feel understandable.

Not merely animated.

Motion Across Themes

Themes should never alter motion philosophy.

Light Mode.

↓

Calm movement.

Dark Mode.

↓

Calm movement.

Only Material interpretation changes.

Motion behaviour remains identical.

Motion Across Devices

Desktop.

↓

Pointer interaction.

Phone.

↓

Touch interaction.

Television.

↓

Remote interaction.

Despite differing interaction methods...

Motion should communicate identical behavioural meaning.

Users should recognise the same Motion language everywhere.

Accessibility

Motion should always remain optional.

Reducing Motion should never reduce understanding.

Examples.

Reduced Motion.

↓

Less animation.

↓

Identical hierarchy.

↓

Identical behaviour.

Accessibility should remove movement.

Never remove meaning.

Good Examples

Hero Transition

Current Hero.

↓

Hero gently evolves.

↓

Atmosphere follows.

↓

Supporting information reorganises.

Nothing appears replaced.

Overlay

Overlay emerges.

↓

Interaction.

↓

Overlay retreats.

↓

World continues.

The user's Context remains uninterrupted.

Playback

Video becomes dominant.

↓

Controls appear only when required.

↓

Controls quietly disappear.

↓

Entertainment continues.

Motion supports immersion.

Anti-patterns

Decorative Motion

Movement exists because it looks impressive.

Surprise Motion

Objects move unexpectedly without behavioural justification.

Competitive Motion

Multiple unrelated animations occur simultaneously.

Hierarchy weakens.

Motion As Branding

Movement exists primarily to advertise product identity.

Understanding decreases.

Motion Philosophy Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> Understanding
Understanding --> Motion
Motion --> Materials
Materials --> Presentation
Presentation
Presentation --> UserPerception
```

Motion exists because behaviour changed.

Never because animation is available.

Relationship To Future Chapters

The remaining chapters define how this philosophy becomes implementation.

Including:

- Motion Hierarchy
- Behavioural Motion
- Material Motion
- Refraction Motion
- Temporal Continuity
- Motion Curves
- Accessibility
- Runtime Motion Resolution

Every implementation should reinforce the philosophy established here.

Summary

Motion is not visual decoration.

It is behavioural communication.

Users should never admire the animation.

They should simply understand what happened.

When Motion succeeds, the interface feels alive without ever feeling busy.

That quiet clarity is the defining characteristic of the Mosaic Motion System.

Review Status

Status

Draft

Next File

`02-motion-hierarchy.md`

Motion Hierarchy

Purpose

Not every movement deserves equal attention.

Just as Composition organises understanding through hierarchy...

Motion organises change through hierarchy.

Motion Hierarchy determines:

- what moves first,
- what moves most,
- what should barely move at all.

Without hierarchy, movement becomes visual noise.

With hierarchy, movement quietly reinforces understanding.

Definition

Within MDS, **Motion Hierarchy** is defined as:

> ***The ordered system through which behavioural importance determines the visibility, timing and physical presence of movement.***

Motion Hierarchy communicates significance.

It does not create significance.

Behaviour already determines importance.

Motion simply expresses it.

Why Hierarchy Exists

Imagine every element moving simultaneously.

`Hero`

`Navigation`

`Timeline`

`Metadata`

`Artwork`

`Background`

`Buttons`

Nothing feels important.

Everything competes.

Instead.

```
Hero

↓

Supporting Information

↓

Peripheral Information
```

Movement naturally guides attention.

Users immediately understand where behavioural change occurred.

Behaviour Before Motion

Motion Hierarchy always follows Behaviour.

Conceptually.

```
[text]
Behaviour

↓

Composition

↓

Hierarchy

↓

Motion

↓

Presentation
```

Movement should never invent hierarchy.

It should reveal hierarchy that already exists.

Motion Levels

The Motion System defines five conceptual levels.

```
[text]
Hero

↓

Primary

↓

Supporting

↓

Peripheral

↓
```

`Environmental`

Each level communicates a different behavioural importance.

Hero Motion

Purpose.

Communicate major behavioural transitions.

Examples.

- Focus changes
- Hero evolution
- Playback beginning
- Playback ending

Hero Motion receives:

- greatest visual attention,
- longest perceived continuity,
- richest material behaviour.

Hero Motion should remain calm.

Never theatrical.

Primary Motion

Purpose.

Communicate immediate supporting changes.

Examples.

- Continue Watching
- Progress
- Timeline
- Reading Position

Primary Motion follows Hero Motion naturally.

It should never compete with it.

Supporting Motion

Purpose.

Communicate contextual evolution.

Examples.

- Cast
- Reviews
- Metadata
- Related Works

Supporting Motion should remain subtle.

Readers should perceive continuity without consciously following movement.

Peripheral Motion

Purpose.

Communicate background adaptation.

Examples.

- Collections
- Recommendations
- History

Peripheral Motion should be restrained.

Users should notice it only when looking directly at it.

Environmental Motion

Purpose.

Communicate changes in the surrounding environment.

Examples.

- Runtime Atmosphere
- Acrylic lighting
- Refraction
- Canvas

Environmental Motion should rarely attract conscious attention.

Instead...

Users should simply feel that the environment evolved naturally.

Motion Order

Motion should generally follow this sequence.

```
[text]
Behaviour
```

↓

```
Hero

↓

Primary

↓

Supporting

↓

Peripheral

↓

Environment Settles
```

The interface should feel as though understanding propagates naturally through the World.

Hierarchy Is Temporal

Hierarchy influences timing.

Hero Motion begins first.

Supporting Motion follows.

Environmental Motion settles last.

This staggered behaviour reinforces:

- causality,
- continuity,
- physicality.

Movement should resemble consequence rather than choreography.

Hierarchy Is Physical

Motion should respect Material Hierarchy.

Hero Material.

↓

Moves first.

Surface.

↓

Responds.

Canvas.

↓

Barely changes.

Physical behaviour should reinforce perceived material depth.

The environment should feel believable.

Hierarchy Across Domains

The Motion Hierarchy remains identical across every entertainment domain.

Television.

↓

Episode becomes Hero.

Books.

↓

Current Chapter becomes Hero.

Music.

↓

Current Track becomes Hero.

Different content.

One Motion language.

Runtime Hierarchy

Runtime systems should resolve Motion Hierarchy automatically.

Components should never decide:

- movement priority,
- animation order,
- timing relationships.

Applications simply communicate behavioural change.

The Motion System determines everything else.

Accessibility

Reduced Motion should preserve hierarchy.

Example.

Instead of:

```
Hero Moves
```

↓

```
Timeline Moves
```

↓

`Environment Moves`

Use.

`Hero Appears`

↓

`Timeline Updates`

↓

`Environment Quietly Changes`

The amount of movement changes.

The order of understanding does not.

Good Examples

Hero Change

Hero evolves.

↓

Primary information follows.

↓

Supporting information settles.

↓

Atmosphere completes.

Users understand exactly what changed.

Playback

Video becomes dominant.

↓

Controls quietly respond.

↓

Environment stabilises.

Nothing distracts from entertainment.

Reading

Chapter changes.

↓

Progress updates.

↓

Bookmarks settle.

↓

Ambient atmosphere gently adapts.

The rhythm remains calm.

Anti-patterns

Simultaneous Motion

Everything moves together.

Hierarchy disappears.

Decorative Priority

Visually interesting objects move before behaviourally important ones.

Understanding weakens.

Background Dominance

Environmental movement becomes more noticeable than interaction.

Atmosphere competes with content.

Random Ordering

Movement order changes between different parts of the platform.

Users lose predictability.

Motion Hierarchy Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> HeroMotion
HeroMotion --> PrimaryMotion
PrimaryMotion --> SupportingMotion
SupportingMotion --> PeripheralMotion
PeripheralMotion --> EnvironmentalMotion
EnvironmentalMotion --> Presentation
```

Behaviour establishes importance.

Motion communicates it over time.

Relationship To Future Chapters

The next chapter defines **Behavioural Motion**.

Motion Hierarchy answers:

> ***What should move first?***

Behavioural Motion answers:

> ***How should different kinds of behavioural change be communicated?***

Together they establish one coherent language of movement across the entire Mosaic platform.

Summary

Motion Hierarchy transforms behavioural importance into temporal understanding.

Users should instinctively perceive:

- what changed,
- where it changed,
- why it mattered,

without consciously analysing animation.

When Motion Hierarchy succeeds, movement quietly guides attention exactly where understanding requires it.

Review Status

Status

Draft

Next File

`03-behavioural-motion.md`

Behavioural Motion

Purpose

Motion exists because behaviour changes.

Not because interfaces need animation.

This chapter defines the relationship between behavioural events and the movement used to communicate them.

Every meaningful motion within Mosaic should originate from a behavioural change already defined by the Interaction Model.

If no behavioural change occurred...

No motion should occur.

Definition

Within MDS, **Behavioural Motion** is defined as:

> ***The visible expression of behavioural change through controlled physical movement.***

Behavioural Motion answers three questions.

- What changed?
- Why did it change?
- What should the user understand next?

It intentionally avoids answering:

- Which animation should play?

Behaviour Drives Motion

Motion should never be triggered directly by interaction.

Incorrect.

```
[text]
Tap

↓

Animation
```

Correct.

```
[text]
Tap

↓

Behaviour Changes
```

```
↓

Composition Evolves

↓

Motion Explains

↓

Presentation
```

Motion should always describe behaviour.

Never input.

Behaviour Categories

The Motion System recognises several categories of behavioural change.

```
[text]
Focus

↓

Context

↓

Composition

↓

Information

↓

Material

↓

Environment
```

Each category produces different movement.

Focus Motion

Purpose.

Communicate a change in primary attention.

Examples.

- selecting new media
- opening a new Hero
- changing Domains

Characteristics.

- strongest motion
- greatest continuity

- highest editorial importance

Focus Motion should feel deliberate.

Not dramatic.

Context Motion

Purpose.

Communicate changing activity while preserving Focus.

Examples.

- playback begins
- playback pauses
- reading resumes
- search opens

Context Motion should generally feel lighter than Focus Motion.

Users remain inside the same World.

Composition Motion

Purpose.

Communicate changes in hierarchy.

Examples.

- Timeline promoted
- Supporting information reordered
- Hero expands
- Relationships appear

Composition Motion should reinforce understanding.

Not redraw layouts.

Information Motion

Purpose.

Communicate changes to information itself.

Examples.

- progress updates
- downloads complete
- metadata refreshes

- episode released

Information Motion should remain highly local.

Only affected information should move.

Material Motion

Purpose.

Communicate physical response.

Examples.

- Acrylic settles
- Refraction redistributes
- Overlay emerges
- Hero illumination shifts

Material Motion reinforces physicality.

It should never become the centre of attention.

Environmental Motion

Purpose.

Communicate environmental evolution.

Examples.

- Runtime Atmosphere
- Canvas adaptation
- ambient lighting

Environmental Motion should almost disappear into peripheral vision.

Users should feel it more than observe it.

Behavioural Continuity

Every behavioural transition should preserve identity.

Poor.

```
[text]
Old Hero

↓

Removed

↓

New Hero
```

Preferred.

```text
Hero Evolves

↓

Understanding Continues
```

Movement should communicate transformation.

Not replacement.

Locality

Behavioural Motion should remain local whenever possible.

Example.

Progress updates.

Only:

- progress indicator
- remaining time
- nearby timeline

should respond.

The rest of the interface should remain calm.

This significantly reduces cognitive load.

Cause Before Effect

Movement should always respect causality.

Preferred.

```text
Focus Changes

↓

Hero Moves

↓

Supporting Information Responds

↓

Environment Settles
```

Avoid.

```text
Everything Moves

↓

User Works Out What Happened
```

Understanding should precede interpretation.

Behavioural Weight

Different behavioural events possess different conceptual weight.

Examples.

`Pause Playback`

↓

Light Motion.

`Change Hero`

↓

Medium Motion.

`Change Domain`

↓

Strong Motion.

Motion intensity should reflect behavioural significance rather than visual distance.

Behavioural Persistence

Movement should never interrupt ongoing understanding.

Example.

```
Reading

↓

Search

↓

Close Search

↓

Reading
```

Search Motion should preserve reading continuity.

The reader should feel that the original activity remained alive beneath the interaction.

Behaviour Across Domains

Different entertainment domains share the same behavioural language.

Television.

↓

Episode changes.

Books.

↓

Chapter changes.

Music.

↓

Track changes.

Different media.

Identical behavioural categories.

This consistency strengthens the Mental Model.

Runtime Behaviour

The Runtime Motion Resolver should receive behavioural events.

Not interface events.

Conceptually.

```
[text]
Behaviour

↓

Motion Resolver

↓

Material Motion

↓

Presentation
```

Components should never decide:

- timing
- sequencing
- hierarchy

They simply communicate behavioural change.

Accessibility

Reduced Motion should preserve behavioural communication.

Instead of:

```
[text]
Movement
```

Use:

- opacity
- hierarchy
- spacing
- composition

Understanding should remain identical.

Only physical movement reduces.

Good Examples

Hero Change

Focus changes.

↓

Hero evolves.

↓

Supporting information follows.

↓

Atmosphere settles.

Behaviour remains immediately understandable.

Playback

Playback begins.

↓

Controls reduce.

↓

Video becomes Hero.

↓

Environment calms.

The transition feels inevitable.

Reading

Chapter advances.

↓

Progress updates.

↓

Bookmarks settle.

↓

Reader continues naturally.

Nothing feels interrupted.

Anti-patterns

Animation Libraries

Movement selected because an animation exists.

Decorative Motion

Movement exists without behavioural meaning.

Interface Motion

Components moving because layouts changed rather than behaviour.

Behaviour Replacement

Motion replacing behavioural clarity rather than communicating it.

Behavioural Motion Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> MotionCategory
MotionCategory --> MotionHierarchy
MotionHierarchy --> MaterialResponse
MaterialResponse --> Presentation
Presentation --> Understanding
```

Behaviour always precedes motion.

Understanding always follows it.

Relationship To Future Chapters

The next chapter defines **Material Motion**.

Behavioural Motion explains:

> ***Why movement occurs.***

Material Motion explains:

> ***How physical materials participate in that movement.***

Together they establish one coherent physical language of change throughout Mosaic.

Summary

Behavioural Motion transforms behavioural change into understandable physical evolution.

Users should never wonder:

- what happened,
- why it happened,
- what changed.

Movement should quietly answer those questions before they are consciously asked.

That clarity is the defining objective of Behavioural Motion.

Review Status

Status

Draft

Next File

`04-material-motion.md`

Material Motion

Purpose

The Material System established that Mosaic materials are physical behaviours rather than decorative effects.

This chapter defines how those materials move.

Material Motion ensures that:

- Acrylic behaves like Acrylic.
- Canvas behaves like Canvas.
- Hero Materials feel physically substantial.
- Overlay Materials emerge naturally.

Movement should reinforce the illusion that every material occupies one coherent physical environment.

Materials should never feel like independently animated interface layers.

Definition

Within MDS, **Material Motion** is defined as:

> ***The physical response of Mosaic materials to behavioural change while preserving the perception of a coherent environment.***

Material Motion communicates:

- physical presence,
- continuity,
- environmental response.

It intentionally avoids decorative animation.

Philosophy

Imagine several sheets of polished acrylic resting on a table.

Move one.

Nearby sheets respond subtly.

The table remains stable.

Nothing teleports.

Nothing jitters.

The environment behaves as one physical system.

Material Motion should create the same impression.

Materials Move Differently

Not every material should move equally.

Each material possesses its own behavioural characteristics.

Material	Motion Behaviour
Canvas	Almost static
Surface	Gentle response
Acrylic	Soft physical movement
Hero	Strongest physical response
Overlay	Controlled emergence

Material identity should remain recognisable through movement alone.

Canvas Motion

Canvas represents the environment.

It should rarely move.

Examples.

Allowed.

- atmosphere settling
- subtle luminance evolution

Avoid.

- translation
- scaling
- dramatic movement

The environment should remain behaviourally stable while objects move within it.

Surface Motion

Surfaces group information.

Movement should communicate:

- organisation,
- hierarchy,
- continuity.

Surface Motion should feel:

- restrained,
- deliberate,

- stable.

Surfaces should never bounce or exaggerate movement.

Acrylic Motion

Acrylic possesses physical weight.

Movement should therefore include:

- subtle inertia,
- soft settling,
- restrained elasticity.

The objective is not simulation.

The objective is perceived substance.

Users should feel that Acrylic occupies space.

Hero Motion

Hero Material receives the richest physical behaviour.

Examples.

Focus changes.

↓

Hero glides naturally.

↓

Atmosphere redistributes.

↓

Refraction settles.

↓

Supporting materials respond.

Hero Motion should communicate confidence rather than speed.

The Hero should feel physically important.

Overlay Motion

Overlay Material temporarily interrupts the environment.

Its movement should communicate:

- emergence,
- assistance,

- departure.

Preferred.

```
[text]
Overlay

↓

Emerges

↓

Interaction

↓

Returns
```

Avoid.

```
[text]
Overlay

↓

Instantly Appears

↓

Instantly Disappears
```

Users should perceive Overlay as a physical object briefly entering the environment.

Material Relationships

Materials should influence one another.

Example.

Hero shifts.

↓

Nearby Acrylic responds.

↓

Canvas subtly receives new atmosphere.

↓

Overlay remains largely independent.

Movement therefore propagates through the environment rather than occurring independently.

Environmental Stability

Material Motion should preserve one important illusion.

The environment already existed.

Behaviour changed.

The environment responded.

Users should never feel that the interface rebuilt itself.

Refraction Motion

Refraction should move through materials.

Not with them.

Example.

Hero changes.

↓

Material moves.

↓

Environmental light gradually redistributes.

↓

Edges settle.

↓

Atmosphere stabilises.

Light should appear to possess inertia independent from geometry.

Motion And Composition

Composition determines which materials move.

Example.

Progress updates.

↓

Timeline responds.

↓

Supporting Acrylic subtly updates.

↓

Hero remains stable.

Movement should remain local whenever practical.

Temporal Behaviour

Material Motion should evolve over time.

Preferred.

```
[text]
Behaviour

↓

Material Responds

↓

Environment Settles
```

Avoid.

```
[text]
Behaviour

↓

Everything Immediately Complete
```

Settling communicates physical realism.

It also reinforces behavioural continuity.

Motion Across Themes

Light Theme.

↓

Softer diffusion.

↓

Gentler settling.

Dark Theme.

↓

Richer environmental response.

↓

Slightly deeper perceived momentum.

Theme influences perception.

Not behaviour.

Motion Across Devices

Desktop.

↓

Highest fidelity.

Television.

↓

Greater perceived depth.

Phone.

↓

Simplified motion.

Low Power Device.

↓

Reduced material complexity.

Despite differing implementations...

Materials should always feel like the same physical objects.

Accessibility

Reduced Motion should simplify physical response.

Examples.

Instead of:

- translation,
- layered settling,
- atmospheric redistribution,

prefer:

- opacity,
- hierarchy,
- subtle luminance changes.

The physical language should remain understandable even when movement is reduced.

Runtime Material Motion

The Runtime Motion Resolver should resolve Material Motion according to:

```
[text]
Behaviour

↓

Material Identity

↓

Accessibility

↓

Device Capability

↓

Runtime Motion
```

Components should never manually animate materials.

The Material System owns physical behaviour.

Performance

Future implementations should optimise Material Motion through:

- shared material timelines,
- cached atmospheric interpolation,
- GPU acceleration,
- incremental updates.

Materials should continue feeling premium without compromising responsiveness.

Plugins

Extensions never animate materials.

Plugins contribute:

- behavioural events,
- artwork,
- information.

The Motion System determines:

- material response,
- atmosphere redistribution,
- physical continuity.

Every extension therefore inherits one coherent physical language automatically.

Good Examples

Hero Transition

Hero moves.

↓

Nearby Acrylic follows.

↓

Atmosphere settles.

↓

Canvas remains calm.

The World feels continuous.

Overlay

Overlay rises naturally.

↓

Interaction occurs.

↓

Overlay returns.

↓

Hero immediately regains presence.

No visual competition exists.

Playback

Video becomes dominant.

↓

Overlay controls softly emerge.

↓

Controls disappear.

↓

Environment settles.

Motion supports immersion.

Anti-patterns

Floating Materials

Materials move independently from one another.

Physical coherence disappears.

Decorative Bounce

Elasticity exists purely because it appears playful.

Material credibility weakens.

Static Acrylic

Materials ignore behavioural change.

The environment feels lifeless.

Instant Atmosphere

Environmental lighting changes without physical transition.

Continuity weakens.

Material Motion Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> MaterialIdentity
MaterialIdentity --> MaterialMotion
MaterialMotion --> RefractionResponse
RefractionResponse --> EnvironmentSettles
EnvironmentSettles --> Presentation
```

Behaviour changes first.

Materials respond.

The environment quietly settles afterwards.

Relationship To Future Chapters

The next chapter defines **Refraction Motion**.

Material Motion explains:

> ***How physical materials move.***

Refraction Motion explains:

> ***How light itself moves through those materials over time.***

Together they complete the physical language of movement within Mosaic.

Summary

Material Motion transforms behavioural change into believable physical response.

Materials should feel:

- substantial,
- connected,
- calm,
- consistent.

Users should never perceive animated panels.

They should perceive one coherent environment naturally responding to the evolution of their entertainment World.

Review Status

Status

Draft

Next File

`05-refraction-motion.md`

Refraction Motion

Purpose

The Material System established that Refraction is the transport of environmental light through Mosaic materials.

Material Motion established that physical materials respond to behavioural change.

This chapter defines how **light itself** behaves over time.

Unlike geometry, light possesses inertia.

It spreads.

Settles.

Diffuses.

Refraction Motion exists to communicate that environmental behaviour.

The objective is not visual spectacle.

It is environmental continuity.

Definition

Within MDS, **Refraction Motion** is defined as:

> ***The temporal evolution of environmental light as it propagates through the Mosaic Material System following behavioural change.***

Refraction Motion belongs to:

- atmosphere,
- light,
- diffusion,

rather than:

- geometry,
- layout,
- components.

Philosophy

Imagine changing the lighting within a room.

The walls do not instantly change.

The light gradually redistributes.

Shadows soften.

Reflections move.

The room settles.

Mosaic should create the same perception.

Behaviour changes.

The environment responds.

Light Moves Differently

Materials move physically.

Light moves environmentally.

These are different behaviours.

Conceptually.

```
[text]
Behaviour

↓

Material Motion

↓

Light Redistribution

↓

Environment Settles
```

Users should perceive:

- objects moving,
- light following.

Not both occurring simultaneously.

Refraction Follows Behaviour

Refraction Motion should never occur independently.

Incorrect.

```
[text]
Light Moves

↓

Nothing Changed
```

Correct.

```
[text]
Behaviour Changes

↓

Composition Evolves
```

```
↓

Materials Respond

↓

Light Redistributes
```

Light should always appear to respond.

Never initiate.

Refraction Timeline

Every significant environmental transition should broadly follow the same conceptual sequence.

```
[text]
Behaviour

↓

Material Motion Begins

↓

Refraction Redistributes

↓

Diffusion Softens

↓

Environment Settles
```

Each stage reinforces physical continuity.

Hero Refraction

Hero changes produce the strongest Refraction Motion.

Example.

```
Old Hero

↓

New Hero

↓

Atmosphere Blends

↓

Light Field Reprojects

↓

Hero Acrylic Settles
```

The Hero should appear to illuminate the surrounding environment naturally.

Not instantly repaint it.

Supporting Refraction

Supporting materials receive delayed environmental response.

Example.

Hero changes.

↓

Nearby Timeline.

↓

Related Works.

↓

Peripheral Surfaces.

↓

Canvas.

Light should appear to propagate naturally through the interface.

Canvas Behaviour

Canvas should respond last.

Examples.

Atmosphere changes.

↓

Canvas luminance subtly evolves.

↓

Environmental colour gently settles.

Canvas should never appear animated.

It should feel like the environment itself slowly responding.

Overlay Behaviour

Overlay Materials intentionally reduce Refraction Motion.

Reasons include:

- readability,
- interaction,
- accessibility.

Interaction should never compete with environmental lighting.

While an Overlay is active:

Environment.

↓

Quiet.

Interaction.

↓

Clear.

Directionality

Refraction Motion should preserve directional continuity.

Example.

Hero.

↓

Upper Left.

↓

Light travels diagonally.

↓

Nearby Acrylic responds first.

↓

Distant Materials respond later.

The environment should preserve a believable spatial relationship.

Diffusion Over Time

Diffusion should evolve continuously.

Preferred.

```
[text]
Strong Reflection

↓

Soft Diffusion

↓

Environmental Balance
```

Avoid.

```
[text]
Strong Reflection

↓
```

```
Instant Neutral
```

Users should perceive the atmosphere settling rather than disappearing.

Temporal Weight

Not every behavioural event deserves environmental redistribution.

Examples.

Progress updates.

↓

No environmental change.

Playback pause.

↓

Minimal environmental change.

Hero changes.

↓

Full environmental redistribution.

The Motion System should respect behavioural significance.

Runtime Atmosphere

Runtime Atmosphere should never jump between states.

Preferred.

```
[text]
Old Atmosphere

↓

Blend

↓

Redistribution

↓

New Atmosphere
```

The user should perceive one continuous environment.

Not multiple independent themes.

Motion And Refraction

Material Motion and Refraction Motion intentionally differ.

Material Motion answers:

> *What moved?*

Refraction Motion answers:

> *How did the environment respond?*

These systems should complement one another.

Never duplicate one another.

Accessibility

Reduced Motion should simplify Refraction Motion.

Examples.

Instead of:

- gradual light transport,
- animated diffusion,
- atmospheric settling,

prefer:

- immediate stable lighting,
- subtle luminance adjustment.

The environment should remain coherent without unnecessary movement.

Performance

Future implementations should optimise Refraction Motion aggressively.

Preferred techniques include:

- cached UV fields,
- temporal interpolation,
- GPU blending,
- incremental updates.

Environmental motion should never become computationally dominant.

Plugins

Extensions never participate in Refraction Motion.

Plugins contribute:

- artwork,
- behavioural events.

The Motion System determines:

- light redistribution,
- diffusion,
- temporal behaviour.

Every extension therefore inherits identical environmental motion.

Good Examples

Hero Change

Hero moves.

↓

Materials respond.

↓

Environmental light follows.

↓

Canvas settles.

The world feels physically believable.

Playback

Video becomes Hero.

↓

Controls emerge.

↓

Environment calms.

↓

Playback continues.

Motion quietly supports immersion.

Reading

Chapter changes.

↓

Hero evolves.

↓

Warm atmosphere slowly redistributes.

↓

Reading continues uninterrupted.

The transition feels almost imperceptible.

Anti-patterns

Animated Glow

Glow moves independently from behavioural change.

Flashing Atmosphere

Entire interface instantly changes colour.

Uniform Refraction

Every material receives identical environmental updates.

Decorative Light

Light exists because animation is available.

Not because behaviour changed.

Refraction Motion Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> MaterialMotion
MaterialMotion --> RuntimeAtmosphere
RuntimeAtmosphere --> RefractionMotion
RefractionMotion --> Diffusion
Diffusion --> EnvironmentSettles
EnvironmentSettles --> Presentation
```

Behaviour begins the transition.

Refraction completes the environmental response.

Relationship To Future Chapters

The next chapter defines **Temporal Continuity**.

Refraction Motion explains:

> ***How environmental light evolves.***

Temporal Continuity explains:

> ***How every movement across the entire platform becomes one continuous behavioural experience.***

Together they ensure Mosaic feels like one living environment rather than a collection of animated interfaces.

Summary

Refraction Motion is the movement of environmental light.

It should feel:

- calm,
- physical,
- continuous,
- inevitable.

Users should never consciously follow the light.

They should simply feel that the world around their entertainment naturally responded when something meaningful changed.

That quiet environmental response is one of the defining characteristics of the Mosaic Motion System.

Review Status

Status

Draft

Next File

`06-temporal-continuity.md`

Temporal Continuity

Purpose

Movement is only meaningful when users perceive it as part of one continuous experience.

This chapter defines how Mosaic preserves that perception across time.

Unlike traditional interfaces, which often present interaction as a sequence of unrelated transitions, Mosaic models interaction as the continuous evolution of one persistent World.

Temporal Continuity ensures that users never feel as though they are repeatedly entering new interfaces.

Instead...

Their World simply continues.

Definition

Within MDS, **Temporal Continuity** is defined as:

> ***The preservation of behavioural, material and perceptual continuity throughout every change in the user's World.***

Temporal Continuity is not:

- transition timing,
- animation duration,
- easing.

It is the user's perception that:

> ***Nothing was interrupted.***

Philosophy

Imagine turning a page in a book.

The story does not restart.

The paper changes.

The reader continues.

Temporal Continuity should create the same feeling throughout Mosaic.

Every interaction should feel like:

`Continuation`

↓

`Evolution`

↓

```
Continuation
```

Never:

```
Stop

↓

Replace

↓

Restart
```

Why Continuity Exists

Entertainment is inherently continuous.

Examples.

- watching a series,
- reading a novel,
- listening to an album.

Traditional interfaces frequently interrupt that continuity with:

- page changes,
- loading states,
- unrelated transitions,
- promotional interruptions.

Mosaic intentionally avoids these behaviours.

Behaviour Before Time

Time should always follow behaviour.

Conceptually.

```
[text]
Behaviour

↓

Motion

↓

Temporal Continuity

↓

Understanding
```

Animation timing should never become the primary design decision.

Behaviour determines the transition.

Time simply communicates it.

Continuity Chain

Every meaningful interaction should preserve:

```
[text]
World

↓

Focus

↓

Context

↓

Composition

↓

Materials

↓

Typography

↓

Motion

↓

Understanding
```

Breaking any link weakens the user's perception of continuity.

Progressive Evolution

Changes should unfold progressively.

Preferred.

```
[text]
Old World

↓

Behaviour Changes

↓

Materials Respond

↓

Typography Adjusts

↓

Environment Settles

↓
```

`New World`

Avoid.

```
[text]
Old World
```

↓

`Everything Replaced`

↓

`New World`

Users should perceive evolution.

Not reconstruction.

Object Permanence

Objects should remain recognisable while changing.

Examples.

`Current Episode`

↓

`Next Episode`

The object evolves.

It should not disappear and be replaced by another unrelated interface.

Maintaining object permanence significantly reduces cognitive effort.

Spatial Continuity

Users should maintain an intuitive sense of place.

Motion should preserve:

- direction,
- origin,
- destination,
- relationships.

Objects should never appear to teleport without behavioural justification.

Spatial understanding reinforces temporal understanding.

Environmental Continuity

The environment should remain remarkably stable.

Examples.

Canvas.

↓

Stable.

Atmosphere.

↓

Gradual evolution.

Hero.

↓

Behavioural response.

Users should feel that:

The world remained.

Only their current experience changed.

Reading Continuity

Typography should remain readable throughout transitions.

Readers should never lose:

- paragraph rhythm,
- hierarchy,
- orientation,

because movement occurred.

Editorial continuity remains just as important as spatial continuity.

Material Continuity

Materials should preserve their identity.

Hero Material.

↓

Hero Material.

Overlay.

↓

Overlay.

Canvas.

↓

Canvas.

Materials may evolve.

They should never unexpectedly become different materials.

The physical world should remain internally consistent.

Context Preservation

Many behavioural changes alter Context without changing Focus.

Examples.

Playback.

↓

Pause.

↓

Subtitles.

↓

Playback.

Temporal Continuity should preserve the user's understanding that:

The same experience continues.

Multi-Step Behaviour

Complex interactions should feel like one transition.

Example.

```
Select Episode

↓

Hero Evolves

↓

Timeline Updates

↓

Atmosphere Settles

↓

Playback Begins
```

Users should perceive:

One evolving experience.

Not four independent animations.

Time And Refraction

Environmental light should continue evolving after physical movement has largely completed.

Conceptually.

```
[text]
Material Moves

↓

Refraction Redistributes

↓

Environment Settles
```

This slight temporal offset strengthens perceived physicality.

Time And Atmosphere

Runtime Atmosphere should blend.

Never switch.

Preferred.

```
[text]
Old Atmosphere

↓

Blend

↓

New Atmosphere
```

Avoid.

```
[text]
Old Atmosphere

↓

Replace

↓

New Atmosphere
```

Atmosphere should feel environmental.

Not procedural.

Interruptions

Temporary interruptions should preserve continuity.

Examples.

Phone notification.

↓

Return.

Search.

↓

Return.

Settings.

↓

Return.

Users should always feel they returned to the same World.

Accessibility

Reduced Motion should preserve Temporal Continuity.

Movement may reduce.

Continuity must remain.

Examples include:

- opacity transitions,
- hierarchy updates,
- material changes,
- typography stability.

Understanding should survive even when movement does not.

Runtime Behaviour

The Runtime Motion Resolver should evaluate continuity globally.

Rather than asking:

> *Which animation plays?*

It should ask:

> ***What must remain understandable throughout this behavioural change?***

This shift in perspective defines the Motion System.

Plugins

Extensions should never introduce independent temporal models.

Plugins contribute:

- behaviour,
- information,
- artwork.

The Motion System determines:

- sequencing,
- continuity,
- environmental evolution.

Every extension therefore feels like part of one continuous platform.

Good Examples

Hero Change

Old Hero.

↓

New Hero emerges naturally.

↓

Atmosphere follows.

↓

Reader immediately understands the relationship.

Playback

Overlay appears.

↓

Interaction.

↓

Overlay disappears.

↓

Playback continues.

The experience never feels interrupted.

Reading

Current Chapter.

↓

Next Chapter.

↓

Progress updates.

↓

Environment remains calm.

Reading continues effortlessly.

Anti-patterns

Flash Transition

Entire interface abruptly replaced.

Sequential Animations

Users wait for unrelated movements to complete before continuing.

Reset Behaviour

Small interactions restart the experience.

Independent Systems

Materials, typography and composition evolve independently.

The world fragments.

Temporal Continuity Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> Motion
Motion --> Materials
Materials --> Refraction
Refraction --> Typography
Typography --> Environment
Environment --> Understanding
```

Every system contributes toward one continuous behavioural experience.

Relationship To Future Chapters

The next chapter defines **Motion Curves**.

Temporal Continuity explains:

> ***What should feel continuous.***

Motion Curves explain:

> ***How physical movement should behave while preserving that continuity.***

Together they establish the temporal language of the Mosaic Motion System.

Summary

Temporal Continuity is one of the defining architectural principles of Mosaic.

Users should never feel that they are travelling between disconnected interfaces.

Instead...

Their World should quietly continue evolving around them.

That uninterrupted perception of continuity is ultimately more important than any individual animation.

Review Status

Status

Draft

Next File

`07-motion-curves.md`

Motion Curves

Purpose

Movement possesses character.

Two objects may travel the same distance over the same duration while communicating completely different meanings.

Motion Curves define that character.

Within Mosaic, Motion Curves are **not** aesthetic preferences.

They communicate physical behaviour.

They answer one question.

> ***"How should this movement feel?"***

Not:

> *"Which easing function should we use?"*

Definition

Within MDS, **Motion Curves** are defined as:

> ***The behavioural characteristics governing how movement accelerates, travels and settles over time while preserving the physical language of the Mosaic Material System.***

Curves communicate behaviour.

They do not decorate movement.

Philosophy

Imagine gently placing a book onto a wooden table.

The book does not:

- snap,
- bounce,
- overshoot dramatically.

It slows naturally.

Settles.

Stops.

Mosaic Motion should communicate the same confidence.

Movement should feel:

- deliberate,
- physical,
- calm.

Curves Follow Behaviour

Motion Curves should always be selected according to behavioural intent.

Incorrect.

```
[text]
Every Animation

↓

Same Curve
```

Preferred.

```
[text]
Behaviour

↓

Motion Category

↓

Motion Curve
```

The curve exists because behaviour differs.

Not because movement exists.

Curve Families

The Motion System defines five conceptual curve families.

```
[text]
Emergence

↓

Transition

↓

Settlement

↓

Environment

↓

Instant
```

Each family communicates a different behavioural meaning.

Emergence

Purpose.

Communicate appearance.

Examples.

- Overlay enters
- Hero appears
- Search opens

Characteristics.

- confident acceleration,
- gentle arrival,
- no overshoot.

Emergence should feel intentional rather than theatrical.

Transition

Purpose.

Communicate evolution.

Examples.

- Hero changes
- Composition reorganises
- Context changes

Transition Curves should minimise the feeling of interruption.

Objects should appear to continue their journey rather than restart it.

Settlement

Purpose.

Communicate completion.

Examples.

- Acrylic settles
- Refraction stabilises
- Materials finish responding

Settlement should feel calm.

Not elastic.

Users should perceive confidence rather than softness.

Environment

Purpose.

Communicate environmental adaptation.

Examples.

- Runtime Atmosphere
- Canvas evolution
- Refraction redistribution

Environmental Curves should remain almost imperceptible.

The environment should appear to breathe rather than animate.

Instant

Purpose.

Communicate immediate understanding.

Examples.

- accessibility changes
- reduced motion
- urgent interaction feedback

Instant behaviour intentionally removes unnecessary movement.

The user's understanding remains identical.

Only the transition changes.

Physical Behaviour

Every Motion Curve should suggest:

- inertia,
- momentum,
- restraint,
- physical presence.

Avoid:

- exaggerated bounce,
- rubber-band behaviour,
- cartoon elasticity.

Mosaic materials should feel premium.

Not playful.

Material Curves

Different materials naturally prefer different curves.

Material	Preferred Curve
Canvas	Environment
Surface	Settlement
Acrylic	Transition
Hero	Transition + Settlement
Overlay	Emergence + Settlement

The Material System therefore reinforces the Motion Hierarchy.

Behavioural Weight

Heavier behavioural events should generally feel:

- slower to begin,
- smoother through transition,
- more deliberate when settling.

Lighter behavioural events should feel:

- quicker,
- simpler,
- quieter.

Motion duration alone should never communicate behavioural importance.

The curve contributes equally.

Typography

Typography should move differently from Materials.

Examples.

Preferred.

```text
Typography
```

↓

```
Minimal Movement
```

↓

```
Maximum Readability
```

Avoid.

Large positional movement.

Excessive scaling.

Typography should preserve reading continuity.

Materials should communicate physicality.

Refraction

Refraction should not use the same curve as geometry.

Preferred.

```text
Material Moves

↓

Refraction Lags Slightly

↓

Environment Settles
```

This subtle temporal offset strengthens perceived physical realism.

Runtime Atmosphere

Runtime Atmosphere should evolve using Environmental Curves.

Artwork changes.

↓

Atmosphere slowly blends.

↓

Materials respond.

↓

Environment settles.

Atmosphere should never appear to animate independently.

Accessibility

Reduced Motion should simplify curves.

Preferred.

```
Transition

↓
```

`Minimal`

↓

`Immediate Settlement`

Understanding should remain.

Only the amount of physical movement changes.

Platform Behaviour

Different rendering technologies may implement curves differently.

Web.

↓

CSS timing.

Flutter.

↓

Physics simulation.

SwiftUI.

↓

Native interpolation.

The perceived behaviour should remain recognisably Mosaic.

Performance

Curves should remain computationally inexpensive.

Future implementations should favour:

- deterministic interpolation,
- shared timing profiles,
- predictable execution.

Complex mathematical models should be introduced only when they improve behavioural understanding.

Plugins

Extensions never choose Motion Curves.

Plugins communicate:

- behavioural events.

The Motion System determines:

- hierarchy,

- sequencing,
- curve family,
- timing.

Every extension therefore inherits the same movement language.

Good Examples

Hero

Hero begins confidently.

↓

Moves smoothly.

↓

Settles naturally.

↓

Environment follows.

The transition feels inevitable.

Overlay

Overlay emerges.

↓

Interaction occurs.

↓

Overlay settles.

↓

Environment remains calm.

The user never loses orientation.

Reading

Chapter changes.

↓

Typography remains readable.

↓

Materials respond quietly.

↓

Reading continues.

Movement supports rather than interrupts reading.

Anti-patterns

Bounce Everywhere

Every movement overshoots.

Physical credibility disappears.

Linear Motion

Objects move mechanically.

Nothing feels physically present.

Decorative Curves

Curves selected because they appear fashionable.

Behaviour becomes inconsistent.

Material Competition

Typography, Materials and Atmosphere all use different movement languages.

The interface fragments.

Motion Curve Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> MotionCategory
MotionCategory --> CurveFamily
CurveFamily --> MaterialResponse
MaterialResponse --> Presentation
Presentation --> Understanding
```

Curves communicate behavioural character.

They never define behaviour themselves.

Relationship To Future Chapters

The next chapter defines **Accessibility**.

Motion Curves explain:

> ***How movement should feel.***

Accessibility explains:

> ***How that movement adapts while preserving understanding for every user.***

Together they complete the behavioural language of motion.

Summary

Motion Curves are the emotional cadence of movement.

They should feel:

- calm,
- confident,
- physical,
- inevitable.

Users should never notice the easing function.

They should simply feel that the world moved exactly as they expected it would.

That quiet predictability is the defining characteristic of Mosaic Motion.

Review Status

Status

Draft

Next File

`08-accessibility.md`

Motion Accessibility

Purpose

Motion exists to communicate understanding.

If movement becomes uncomfortable, distracting or inaccessible, it ceases to fulfil that purpose.

Accessibility is therefore not a limitation placed upon the Motion System.

It is one of its primary design objectives.

The Motion System should adapt to people.

People should never be expected to adapt to movement.

Philosophy

The Motion System follows one fundamental rule.

> ***Understanding must survive the complete removal of motion.***

Movement may strengthen understanding.

It must never become the only mechanism communicating:

- hierarchy,
- continuity,
- causality,
- orientation,
- behavioural change.

Reduced Motion should remove animation.

It should never remove meaning.

Accessibility Before Motion

Whenever behavioural understanding conflicts with animation:

Understanding wins.

Always.

Poor.

```text
Beautiful Animation

↓

User Discomfort
```

Preferred.

```
[text]
Clear Behaviour

↓

Optional Motion
```

The purpose of Motion is communication.

Not spectacle.

Behaviour Is Stable

Accessibility should never alter:

- Behaviour,
- Composition,
- Hierarchy,
- Editorial Structure.

Instead it alters:

- movement,
- timing,
- interpolation,
- material response.

The behavioural language of Mosaic should remain identical regardless of motion preference.

Reduced Motion

Reduced Motion should preserve behavioural understanding while reducing physical movement.

Examples.

Instead of:

- translation,
- scaling,
- layered transitions,

prefer:

- opacity,
- hierarchy changes,
- material state changes,
- immediate continuity.

Users should still understand exactly what changed.

Material Accessibility

Materials should simplify naturally.

Examples.

Hero Material.

↓

Reduced physical movement.

Refraction.

↓

Reduced redistribution.

Atmosphere.

↓

Simplified blending.

Canvas.

↓

Almost unchanged.

The Material System should remain recognisably Mosaic while respecting accessibility preferences.

Refraction Accessibility

Refraction Motion should reduce significantly.

Preferred.

```
[text]
Atmosphere

↓

Stable Lighting

↓

Subtle Material Update
```

Avoid.

```
[text]
Moving Refraction

↓

Animated Lighting

↓

Environmental Drift
```

The environment should remain calm.

Overlay Behaviour

Overlays should prioritise:

- clarity,
- orientation,
- predictability.

Reduced Motion should simplify Overlay behaviour.

Preferred.

```
[text]
Overlay Appears

↓

Interaction

↓

Overlay Disappears
```

No unnecessary physical transitions should remain.

Hero Behaviour

The Hero should remain recognisable even when movement is removed.

Examples.

Hero changes.

↓

Editorial hierarchy updates.

↓

Materials update.

↓

Understanding preserved.

Movement is optional.

Behaviour is not.

Temporal Continuity

Temporal Continuity should survive Reduced Motion.

Users should continue feeling that:

The same World continues.

Rather than:

A different interface appeared.

Continuity is conceptual.

Not animated.

Typography

Typography should remain extremely stable.

Avoid:

- animated line movement,
- character transitions,
- decorative text effects.

Preferred.

Stable typography.

↓

Editorial hierarchy updates.

↓

Readers continue naturally.

Reading comfort always possesses higher priority than movement.

Runtime Atmosphere

Atmosphere should simplify rather than disappear.

Preferred.

```
[text]
Reduced Motion

↓

Shorter Blend

↓

Stable Environment
```

The interface should remain emotionally coherent while reducing environmental animation.

Interaction Feedback

Essential interaction feedback should remain.

Examples include:

- focus indication,

- selection,
- activation,
- confirmation.

Removing these entirely weakens usability.

The Motion System should distinguish between:

Decorative movement.

and

Functional feedback.

Only the former should disappear.

Accessibility Profiles

Future implementations may expose conceptual motion profiles.

Examples.

```
[text]
Full Motion

↓

Reduced Motion

↓

Minimal Motion

↓

Instant
```

Each profile preserves:

- hierarchy,
- continuity,
- understanding.

Only physical expression changes.

Runtime Resolution

The Runtime Motion Resolver should evaluate:

```
[text]
Behaviour

↓

Accessibility

↓

Material Identity
```

↓

`Motion Profile`

↓

`Presentation`

Accessibility should always be evaluated before motion curves.

Platform Behaviour

Platforms should honour operating system accessibility preferences automatically.

Examples include:

- iOS Reduce Motion,
- Android Remove Animations,
- Windows Animation Effects,
- macOS Reduce Motion.

Platform behaviour should integrate naturally into the Runtime Motion Resolver.

Applications should not implement platform-specific logic independently.

Plugins

Extensions should never override Motion Accessibility.

Plugins contribute:

- behavioural events.

The Motion System determines:

- movement,
- simplification,
- continuity.

Every extension therefore automatically inherits accessible motion.

Good Examples

Hero Change

Reduced Motion.

↓

Hero updates.

↓

Materials settle immediately.

↓

Reader understands the new Focus.

Nothing essential has been lost.

Playback

Overlay appears.

↓

Playback continues.

↓

Overlay disappears.

Interaction remains clear.

Motion remains minimal.

Reading

Chapter changes.

↓

Editorial hierarchy updates.

↓

Reader continues naturally.

Movement never interrupts comprehension.

Anti-patterns

Accessibility Theme

Creating an entirely different motion language.

No Feedback

Removing all movement including functional interaction feedback.

Decorative Accessibility

Retaining decorative motion while removing useful motion.

Platform Divergence

Different clients implementing incompatible Reduced Motion behaviour.

Motion Accessibility Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> AccessibilityProfile
AccessibilityProfile --> MotionResolver
MotionResolver --> MaterialMotion
MaterialMotion --> Presentation
Presentation --> Understanding
```

Accessibility refines movement.

It never changes behaviour.

Relationship To Future Chapters

The next chapter defines **Runtime Motion Resolution**.

Motion Accessibility explains:

> ***How movement adapts to people.***

Runtime Motion Resolution explains:

> ***How every behavioural event becomes the correct motion profile at runtime.***

Together they complete the adaptive architecture of the Motion System.

Summary

Motion Accessibility ensures that every user experiences the same:

- understanding,
- continuity,
- hierarchy,
- Companion,

regardless of how much movement they prefer.

Movement should always remain optional.

Understanding never should.

Review Status

Status

Draft

Next File

`09-runtime-motion-resolution.md`

Runtime Motion Resolution

Purpose

Previous chapters defined:

- Behavioural Motion
- Material Motion
- Refraction Motion
- Temporal Continuity
- Motion Curves
- Motion Accessibility

This chapter defines how those concepts become concrete movement at runtime.

Runtime Motion Resolution is the mechanism that transforms behavioural change into consistent physical motion across every Mosaic client.

Applications should never determine:

- animation timing,
- easing,
- sequencing,
- choreography,
- material response.

Instead they communicate behaviour.

The Motion System resolves everything else.

Definition

Within MDS, **Runtime Motion Resolution** is defined as:

> ***The deterministic process through which behavioural change is transformed into platform-specific motion while preserving continuity, hierarchy and accessibility.***

Runtime Motion Resolution never changes behaviour.

It only determines how that behaviour becomes visible.

Why Resolution Exists

Without Runtime Motion Resolution, every component would need to understand:

- behavioural priority,
- animation timing,

- accessibility,
- motion profiles,
- material response,
- platform capabilities.

Instead.

```
[text]
Behaviour

↓

Runtime Motion Resolver

↓

Resolved Motion

↓

Presentation
```

Components remain behaviourally simple.

The Motion System remains globally consistent.

Resolution Pipeline

Every behavioural event follows the same conceptual pipeline.

```
[text]
Behaviour

↓

Motion Hierarchy

↓

Material Identity

↓

Accessibility

↓

Motion Curves

↓

Platform Capability

↓

Resolved Motion
```

Each stage contributes one responsibility.

Resolution Inputs

The Runtime Motion Resolver evaluates:

```
[text]
Behaviour

↓

Composition

↓

Current Context

↓

Material Identity

↓

Accessibility

↓

Device Capability

↓

Platform

↓

Performance Profile
```

No individual input should override behavioural intent.

Behaviour remains the highest authority.

Resolution Order

Motion should always resolve using the same conceptual order.

```
[text]
1.

Behaviour

↓

2.

Motion Hierarchy

↓

3.

Material Response

↓

4.

Accessibility

↓
```

```
5.

Motion Curves

↓

6.

Platform Adaptation

↓

7.

Resolved Motion
```

Meaning always precedes implementation.

Behavioural Stability

One of the strongest guarantees within Mosaic is:

```
[text]
Behaviour

↓

Always Behaviour
```

Playback remains Playback.

Hero remains Hero.

Overlay remains Overlay.

Runtime Resolution changes only:

- timing,
- interpolation,
- sequencing,
- material response.

Behaviour never changes.

Runtime Context

Motion may adapt according to current activity.

Examples.

Watching.

↓

Minimal movement.

Browsing.

↓

Greater compositional evolution.

Reading.

↓

Editorial pacing.

Administration.

↓

Efficient transitions.

The Motion Resolver selects appropriate physical behaviour while preserving one behavioural language.

Motion Profiles

Future implementations may internally resolve Motion Profiles.

Conceptually.

```
[text]
Hero Transition

↓

Motion Profile

↓

Hierarchy

↓

Material Motion

↓

Refraction

↓

Timing

↓

Resolved Motion
```

Components consume only the completed profile.

Runtime Caching

Resolved Motion Profiles should be reusable.

Typical invalidation events include:

- accessibility changes,
- platform changes,
- performance profile changes,

- Motion System updates.

Ordinary interaction should reuse existing behavioural profiles whenever practical.

Incremental Motion

Motion should resolve incrementally.

Preferred.

```
[text]
Behaviour Changes

↓

Affected Materials Move

↓

Environment Responds
```

Avoid.

```
[text]
Behaviour Changes

↓

Entire Interface Re-animates
```

Incremental movement preserves continuity while reducing unnecessary computation.

Platform Adaptation

Different platforms possess different motion capabilities.

Desktop.

↓

Full Material Motion.

Phone.

↓

Reduced complexity.

Television.

↓

Longer perceived distance.

Reduced Motion.

↓

Simplified transitions.

Despite implementation differences...

Users should perceive identical behavioural communication.

Material Awareness

Motion should respect Material Identity.

Examples.

Hero.

↓

Highest fidelity.

Overlay.

↓

Interaction-first.

Canvas.

↓

Environmental stability.

Material Identity should always influence motion behaviour.

Runtime Atmosphere

Motion may influence how Runtime Atmosphere evolves.

Examples.

Hero changes.

↓

Atmosphere redistributes.

↓

Refraction settles.

Atmosphere should never animate independently from behaviour.

The environment responds because behaviour changed.

Performance Profiles

Future runtime implementations may expose conceptual performance profiles.

Examples.

```
[text]
High
```

↓

```
Full Material Behaviour
```

```
[text]
```

```
Balanced

↓

Reduced Environmental Motion

[text]
Efficiency

↓

Simplified Material Motion
```

Every profile should preserve behavioural understanding.

Only rendering complexity changes.

Deterministic Behaviour

Given identical behavioural inputs...

The Runtime Motion Resolver should always produce identical motion.

This determinism improves:

- testing,
- predictability,
- caching,
- cross-platform consistency.

Users gradually learn the movement language because it remains consistent.

Plugins

Extensions never resolve motion.

Plugins contribute:

- behavioural events,
- information,
- artwork.

The Motion Resolver determines:

- hierarchy,
- sequencing,
- material response,
- timing,
- accessibility.

Every extension therefore inherits one coherent movement language automatically.

Good Examples

Hero Transition

Behaviour.

↓

Focus changes.

↓

Motion Resolver.

↓

Hero evolves.

↓

Atmosphere settles.

↓

Understanding preserved.

Playback

Playback begins.

↓

Overlay retracts.

↓

Video becomes Hero.

↓

Environment stabilises.

Motion reinforces immersion.

Reading

Chapter advances.

↓

Typography remains stable.

↓

Materials respond quietly.

↓

Reader continues naturally.

Anti-patterns

Component Animations

Components independently selecting transitions.

Platform Motion

Every platform inventing different behavioural timing.

Runtime Behaviour

Runtime modifying behavioural meaning.

Full Re-animation

Entire interface animates for local behavioural changes.

Runtime Motion Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> MotionHierarchy
MotionHierarchy --> MaterialResponse
MaterialResponse --> Accessibility
Accessibility --> PlatformAdapter
PlatformAdapter --> ResolvedMotion
ResolvedMotion --> Presentation
```

Applications communicate behaviour.

The Motion System resolves movement.

Relationship To Future Chapter

The next chapter defines **Platform Motion**.

Runtime Motion Resolution explains:

> ***How behaviour becomes motion.***

Platform Motion explains:

> ***How different rendering technologies faithfully express that motion while preserving one behavioural language.***

Together they complete the runtime architecture of the Motion System.

Summary

Runtime Motion Resolution transforms behavioural intent into consistent physical movement.

It preserves:

- hierarchy,
- continuity,
- accessibility,
- material behaviour,
- cross-platform consistency.

Applications should never ask:

> ***"Which animation should I play?"***

They should simply communicate:

> ***Behaviour changed.***

The Motion System does everything else.

Review Status

Status

Draft

Next File

`10-platform-motion.md`

Platform Motion

Purpose

The Motion System defines one behavioural language.

Every Mosaic client is responsible for expressing that language through different rendering technologies.

Platform Motion ensures that:

- Web
- Flutter
- SwiftUI
- Jetpack Compose
- Desktop
- Television
- Future platforms

all communicate identical behavioural understanding despite differing animation APIs and rendering capabilities.

Platform implementation should never redefine motion.

It should simply express it faithfully.

Definition

Within MDS, **Platform Motion** is defined as:

> ***The platform-specific implementation of the Mosaic Motion System while preserving one consistent behavioural language.***

Platform Motion concerns implementation.

It does not concern behavioural meaning.

Philosophy

Different rendering engines possess different strengths.

Examples include:

- CSS animations
- Flutter Impeller
- SwiftUI Animation
- Jetpack Compose Animation
- Native television frameworks

These differences are implementation concerns.

Users should never perceive different motion languages because they changed devices.

The Companion should move identically everywhere.

Platform Independence

Behavioural meaning should remain platform independent.

Conceptually.

```
[text]
Hero Transition

↓

Web

Hero Transition

↓

Flutter

Hero Transition

↓

SwiftUI

Hero Transition

↓

Compose
```

The implementation changes.

The behavioural communication remains identical.

Platform Responsibilities

Each platform implementation is responsible for:

- rendering movement
- timing interpolation
- frame scheduling
- animation APIs
- GPU integration
- performance optimisation

Platforms are **not** responsible for:

- behavioural sequencing
- motion hierarchy

- accessibility policy
- material choreography

Those responsibilities belong to the Motion System.

Motion Resolver

Every platform should consume the output of the Runtime Motion Resolver.

Conceptually.

```
[text]
Behaviour

↓

Motion Resolver

↓

Platform Motion

↓

Rendering
```

Platform implementations should never reinterpret behavioural events.

Frame Consistency

Platform Motion should prioritise:

- smoothness
- determinism
- consistency
- responsiveness

before introducing platform-specific visual enhancements.

Behaviour always possesses higher priority than rendering sophistication.

Material Motion

Every platform should preserve Material Motion.

Examples.

Hero.

↓

Physical transition.

Overlay.

↓

Controlled emergence.

Canvas.

↓

Environmental stability.

Platforms may implement these differently.

Users should perceive identical material behaviour.

Refraction Motion

Platforms supporting advanced rendering may implement:

- GPU refraction
- shader-based diffusion
- dynamic light transport

Platforms without these capabilities should approximate equivalent behaviour.

The physical language should remain recognisable even when implementation differs.

Typography Motion

Typography should remain comparatively stable across every platform.

Platform implementations should avoid:

- unnecessary glyph animation
- decorative text transforms
- platform-specific typography effects

Reading comfort remains the primary objective.

Responsive Motion

Different platforms naturally possess different interaction characteristics.

Phone.

↓

Short interactions.

Desktop.

↓

Pointer precision.

Television.

↓

Remote navigation.

Responsive Motion should adapt:

- physical implementation,
- travel distance,
- responsiveness.

It should never alter behavioural sequencing.

Platform Performance

Different hardware possesses different performance budgets.

Examples.

Desktop GPU.

↓

Highest fidelity.

Phone.

↓

Balanced fidelity.

Low-power device.

↓

Simplified environmental motion.

Performance optimisation should reduce rendering complexity before reducing behavioural clarity.

Accessibility

Platform Motion should automatically respect:

- operating system Reduce Motion
- platform animation preferences
- accessibility APIs

Applications should not implement independent accessibility behaviour.

Platform capabilities should integrate directly into Runtime Motion Resolution.

Synchronisation

Future Mosaic clients may render multiple motion systems simultaneously.

Examples.

- Materials

- Typography
- Atmosphere
- Refraction

Platforms should synchronise these systems so users perceive one coherent behavioural transition rather than independent animations.

Timing Fidelity

Exact animation durations are less important than behavioural timing relationships.

For example.

Hero.

↓

Moves first.

↓

Supporting Materials.

↓

Respond.

↓

Environment settles.

Maintaining this ordering is more important than matching millisecond durations across every platform.

Runtime Atmosphere

Platform implementations should preserve atmospheric continuity.

Examples.

Artwork changes.

↓

Atmosphere blends.

↓

Materials respond.

↓

Refraction settles.

Platforms should avoid abrupt environmental changes even if implementation techniques differ.

International Behaviour

Platform Motion should remain culturally neutral.

Movement should communicate behaviour through:

- continuity
- physicality
- hierarchy

rather than culturally specific animation styles.

The Motion System should remain understandable regardless of locale.

Plugins

Extensions should never provide platform-specific animations.

Plugins contribute:

- behaviour
- information
- artwork

Platform Motion determines:

- implementation
- interpolation
- rendering

Every extension therefore automatically inherits future motion improvements.

Good Examples

Web

CSS or WebGPU.

↓

Runtime Motion Resolver.

↓

Behaviour preserved.

Readers perceive identical continuity.

Flutter

Impeller.

↓

Material Motion.

↓

Refraction.

↓

Consistent behavioural language.

Television

Longer travel.

↓

Greater physical scale.

↓

Identical sequencing.

Users continue feeling the same Companion.

Anti-patterns

Platform Personality

Each client inventing independent movement.

Decorative APIs

Platform-specific effects replacing behavioural clarity.

Performance Behaviour

Reducing behavioural hierarchy to improve frame rate.

Independent Animation

Components animating without Runtime Motion Resolution.

Platform Motion Model

```
[mermaid]
flowchart TD

Behaviour
Behaviour --> RuntimeMotionResolver
RuntimeMotionResolver --> PlatformAdapter
PlatformAdapter --> Web
PlatformAdapter --> Flutter
PlatformAdapter --> SwiftUI
PlatformAdapter --> Compose
PlatformAdapter --> Television
Web --> Presentation
Flutter --> Presentation
SwiftUI --> Presentation
Compose --> Presentation
Television --> Presentation
```

One behavioural language.

Many rendering implementations.

Relationship To Future Chapter

The next chapter defines **Motion System Governance**.

Platform Motion explains:

> ***How platforms implement movement.***

Governance explains:

> ***How that movement remains recognisably Mosaic as the platform evolves over many years.***

Together they complete the implementation architecture of the Motion System.

Summary

Platform Motion ensures that Mosaic moves with one behavioural language across every rendering technology.

Rendering engines may evolve.

Animation APIs may disappear.

New platforms will inevitably emerge.

Users should continue feeling that:

The same Companion.

The same World.

The same calm continuity.

That consistency is one of the defining objectives of the Mosaic Motion System.

Review Status

Status

Draft

Next File

`11-governance.md`

Motion System Governance

Purpose

Motion is one of the most memorable characteristics of a digital product.

Users may never consciously remember:

- easing curves,
- durations,
- interpolation,
- animation APIs.

They will remember:

- whether the interface felt calm,
- whether movement made sense,
- whether the application felt effortless to use.

The Motion System therefore becomes part of the long-term behavioural identity of Mosaic.

This chapter defines how Motion should evolve while preserving that identity.

Governance Philosophy

Motion should evolve technically.

Its behavioural language should remain remarkably stable.

The objective is not preserving animation.

It is preserving:

- behavioural continuity,
- physical consistency,
- calmness,
- predictability.

Rendering technologies will change.

The user's understanding should not.

Motion Is Behaviour

Within Mosaic, Motion is considered behavioural architecture.

Changing Motion changes how users understand the platform.

Consequently, Motion should never be treated as visual polish.

Changes affecting:

- Hero transitions,
- Material Motion,
- Motion Hierarchy,
- Temporal Continuity,

should receive architectural review.

Stable Responsibilities

The following concepts should remain highly stable.

- Motion Philosophy
- Motion Hierarchy
- Behavioural Motion
- Material Motion
- Temporal Continuity
- Accessibility behaviour

These concepts define the behavioural identity of Mosaic.

Evolvable Responsibilities

The following may evolve more frequently.

- animation APIs
- rendering engines
- interpolation algorithms
- GPU pipelines
- performance optimisation
- platform-specific implementation

Implementation should evolve.

Behaviour should remain stable.

Motion Ownership

Motion responsibilities are intentionally separated.

Layer	Owner
Motion Philosophy	Design Systems
Behavioural Motion	Design Systems

Layer	Owner
Runtime Motion Resolver	Runtime Platform
Platform Motion	Client Platforms
Rendering Implementation	Rendering Layer

Ownership protects behavioural consistency while allowing implementation to improve independently.

Introducing New Motion

Before introducing a new movement ask:

Question One

Does behaviour actually require movement?

Question Two

Could existing Motion Hierarchy already communicate this?

Question Three

Does this improve understanding...

or simply increase animation?

Question Four

Would users naturally predict this movement?

Question Five

Will this still feel appropriate after future rendering technologies evolve?

If uncertainty remains...

The proposal should be refined before implementation.

Motion Drift

Motion Drift occurs when:

- different areas move differently,
- platforms invent independent behaviours,
- decorative animation accumulates,
- behavioural sequencing diverges,

- transitions lose predictability.

Motion Drift weakens user confidence because movement gradually stops communicating meaning.

It should therefore be treated as architectural debt.

Motion Debt

Examples include:

- duplicated transitions,
- inconsistent easing,
- component-owned animation,
- decorative choreography,
- undocumented behavioural exceptions.

Motion Debt should be removed continuously.

One coherent behavioural language is more valuable than dozens of impressive transitions.

Runtime Governance

The Runtime Motion Resolver may evolve continuously.

Examples include:

- better scheduling,
- improved interpolation,
- adaptive performance,
- predictive pre-resolution,
- GPU optimisation.

These improvements should require no changes to:

- Components,
- Behaviour,
- Composition,
- Materials.

Applications should remain unaware of runtime improvements.

Accessibility Governance

Accessibility possesses higher authority than animation quality.

No motion proposal should weaken:

- readability,

- predictability,
- orientation,
- comfort.

Movement should always adapt to people.

Never the reverse.

Platform Governance

Every client should communicate one movement language.

Desktop.

↓

Behaviourally identical.

Mobile.

↓

Behaviourally identical.

Television.

↓

Behaviourally identical.

Only implementation changes.

Behaviour must remain recognisably Mosaic.

Plugin Governance

Extensions must never introduce:

- independent animation systems,
- custom transition languages,
- proprietary Material Motion,
- alternative behavioural timing.

Plugins contribute:

- behavioural events,
- information,
- artwork.

The Motion System owns movement.

This guarantees one coherent behavioural experience across the ecosystem.

Review Questions

Every Motion proposal should answer:

- Does this explain behaviour?
- Does this preserve continuity?
- Does this reinforce hierarchy?
- Does this strengthen Material behaviour?
- Would removing this movement reduce understanding?
- Does this still feel unmistakably Mosaic?

If the proposal exists primarily because it "looks smooth", it should be reconsidered.

Validation

Future tooling should automatically validate:

- behavioural sequencing
- Motion Hierarchy
- accessibility profiles
- runtime consistency
- platform parity
- unnecessary motion

Validation should reinforce architectural review.

It should never replace behavioural reasoning.

Governance Workflow

```
[mermaid]
flowchart TD

Proposal
Proposal --> BehaviourReview
BehaviourReview --> ExistingMotion["ExistingMotion?"]
ExistingMotion["ExistingMotion?"] -->|Yes| Reuse
ExistingMotion["ExistingMotion?"] -->|No| ArchitectureReview
ArchitectureReview
ArchitectureReview --> AccessibilityReview
AccessibilityReview --> RuntimeReview
RuntimeReview --> Approval
Approval --> Implementation
```

Refinement should always be preferred over expanding the motion vocabulary.

Success Criteria

The Motion System succeeds when:

- users instinctively understand behavioural change,
- movement remains calm,
- continuity feels effortless,
- accessibility remains uncompromised,
- contributors naturally reuse existing behavioural patterns,
- every Mosaic client moves like the same Companion.

Motion should become invisible.

Only understanding should remain.

Architectural Decisions

ADR	Decision
ADR-134	Motion is treated as behavioural architecture rather than visual decoration.
ADR-135	Behaviour always precedes movement.
ADR-136	Runtime Motion Resolution owns implementation while preserving behavioural meaning.
ADR-137	Accessibility always has higher authority than animation fidelity.
ADR-138	Extensions inherit the Motion System rather than extending it.

Review Status

Status

Draft

Next File

`12-adrs.md`

Architectural Decision Records

Purpose

The Architectural Decision Records (ADRs) contained within MDS-005 preserve the architectural reasoning behind the Mosaic Motion System.

Where previous specifications established:

- Design Tokens
- Colour
- Materials
- Typography

this specification establishes the behavioural language through which the user's World evolves over time.

These ADRs explain why Mosaic deliberately treats motion as behavioural communication rather than visual animation.

Future contributors should consult these records before proposing changes to movement, transitions or runtime motion architecture.

ADR Format

Every Mosaic ADR follows the standard structure.

```
[text]
ADR Number

Status

Context

Decision

Consequences

Alternatives Considered

Related Specifications
```

Each ADR documents one significant architectural decision.

ADR-139

Title

Treat Motion As Behaviour

Status

Accepted

Context

Most interface frameworks define motion as animation.

Founder workshops consistently reinforced that movement should explain behaviour rather than decorate interfaces.

Decision

Motion becomes a behavioural system.

Animation becomes an implementation detail.

Consequences

Future rendering technologies may evolve without changing the behavioural language of Mosaic.

ADR-140

Title

Behaviour Always Precedes Motion

Status

Accepted

Context

Component-driven animation creates inconsistent behavioural communication.

Decision

Behaviour generates Motion.

Motion never generates Behaviour.

Consequences

Users consistently understand:

- what changed,
- why it changed,
- what happens next.

ADR-141

Title

Establish Motion Hierarchy

Status

Accepted

Context

Simultaneous movement weakens attention and increases cognitive effort.

Decision

Movement follows:

- Hero
- Primary
- Supporting
- Peripheral
- Environmental

Consequences

Motion naturally reinforces Composition without requiring additional explanation.

ADR-142

Title

Treat Materials As Participants In Motion

Status

Accepted

Context

Early exploration animated interface geometry independently from materials.

This weakened the illusion of a coherent physical environment.

Decision

Materials become first-class participants within Motion.

Consequences

Hero Materials, Acrylic and Canvas evolve together as one physical system.

ADR-143

Title

Refraction Moves Independently From Geometry

Status

Accepted

Context

Environmental light behaves differently from physical objects.

Decision

Refraction receives its own temporal behaviour.

Geometry moves first.

Environmental light settles afterwards.

Consequences

Materials feel physically illuminated rather than digitally animated.

ADR-144

Title

Temporal Continuity Is A First-Class Architectural Concern

Status

Accepted

Context

Traditional applications frequently interrupt users through unrelated transitions and page replacements.

Decision

The Motion System prioritises preserving one continuous World.

Consequences

Users perceive behavioural evolution rather than interface replacement.

ADR-145

Title

Runtime Motion Owns Implementation

Status

Accepted

Context

Allowing components to define transitions fragments behavioural consistency.

Decision

Components communicate Behaviour.

The Runtime Motion Resolver determines:

- sequencing,
- timing,
- curves,
- material response.

Consequences

Applications remain behaviourally simple while runtime implementation continues evolving.

ADR-146

Title

Accessibility Overrides Motion Fidelity

Status

Accepted

Context

Highly expressive motion systems frequently reduce comfort and accessibility.

Decision

Accessibility possesses higher authority than animation quality.

Consequences

Behaviour remains understandable regardless of motion preference.

ADR-147

Title

Platform Motion Implements Rather Than Redefines

Status

Accepted

Context

Different rendering technologies naturally expose different animation capabilities.

Decision

Platforms implement the Motion System.

They do not reinterpret it.

Consequences

Users experience one behavioural language across every Mosaic client.

ADR-148

Title

Extensions Inherit Motion

Status

Accepted

Context

Allowing extensions to introduce independent motion languages fragments behavioural consistency.

Decision

Extensions contribute:

- behaviour,
- information,
- artwork.

The platform owns all movement.

Consequences

Community extensions automatically inherit future Motion System improvements.

ADR Relationships

```
[mermaid]
flowchart TD

ADR139["Behaviour"]

ADR140["Behaviour First"]

ADR141["Hierarchy"]

ADR142["Materials"]

ADR143["Refraction"]

ADR144["Continuity"]

ADR145["Runtime"]

ADR146["Accessibility"]

ADR147["Platforms"]

ADR148["Extensions"]

ADR139 --> ADR140
ADR140 --> ADR141
ADR141 --> ADR142
ADR142 --> ADR143
ADR143 --> ADR144
ADR144 --> ADR145
ADR145 --> ADR146
ADR145 --> ADR147
ADR147 --> ADR148
```

Together these decisions establish one coherent behavioural language for movement throughout the Mosaic platform.

Future ADRs

Future Motion ADRs are expected to formalise:

- Predictive Motion
- AI-assisted Behaviour Resolution
- Shared Multi-User Motion
- Cross-Device Continuation Motion
- HDR Motion Synchronisation
- Haptic Motion Integration
- Spatial Computing Motion
- Adaptive Motion Personas

These intentionally remain outside the scope of MDS-005 Version 0.1.

ADR Governance

Motion ADRs should change only when:

- behavioural research identifies deficiencies,
- accessibility research requires refinement,
- runtime architecture evolves,
- the Design Language itself changes.

New animation technologies alone should never justify architectural changes.

Behaviour should remain recognisably Mosaic regardless of implementation.

Summary

The ADRs contained within MDS-005 define the behavioural identity of Mosaic.

Movement is not treated as visual polish.

It is treated as behavioural communication.

Every implementation should therefore preserve:

- continuity,
- hierarchy,
- physicality,
- calmness,

while allowing animation technologies to evolve independently.

Review Status

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`13-contributor-guidance.md`

Contributor Guidance

Purpose

The Motion System is one of the easiest parts of a Design System to misuse.

Adding movement is simple.

Designing meaningful movement is significantly harder.

This guidance exists to ensure every contributor strengthens the behavioural language of Mosaic rather than adding isolated animations.

The objective is not smoother interfaces.

The objective is clearer understanding.

Think In Behaviour

Never begin with:

> *"What animation should play?"*

Instead ask:

> ***"What behavioural change occurred?"***

Good.

```
[text]
Focus Changed

↓

Motion
```

Poor.

```
[text]
Button Clicked

↓

Animation
```

Motion should always originate from behaviour.

Preserve Continuity

Before introducing movement ask:

> *Does this preserve the user's World...*

or restart it?

Preferred.

```
[text]
```

```
Current State

↓

Evolution

↓

Next State
```

Avoid.

```
[text]
Current State

↓

Replace

↓

Next State
```

Users should feel continuity.

Not replacement.

Respect Motion Hierarchy

Movement should follow the established hierarchy.

```
[text]
Hero

↓

Primary

↓

Supporting

↓

Peripheral

↓

Environment
```

If every object moves equally...

Nothing communicates importance.

Let Materials Move

Components should never animate independently from Materials.

Preferred.

```
[text]
Behaviour

↓
```

```
Material Motion
```

↓

```
Presentation
```

Avoid.

```
[text]
Component
```

↓

```
Animation
```

↓

```
Material Responds Later
```

The Material System owns physical behaviour.

Let Light Follow

Remember the distinction.

Objects move.

Light follows.

Hero changes.

↓

Hero moves.

↓

Atmosphere redistributes.

↓

Environment settles.

Do not animate environmental lighting independently from behavioural change.

Protect Typography

Typography should remain readable throughout movement.

Avoid:

- animated text effects,
- excessive scaling,
- decorative rotation,
- unnecessary opacity flicker.

Readers should never lose editorial continuity because something moved.

Accessibility First

Whenever movement conflicts with comfort:

Comfort wins.

Examples.

Poor.

```
[text]
Beautiful Animation

↓

Motion Sickness
```

Preferred.

```
[text]
Behaviour

↓

Minimal Motion

↓

Same Understanding
```

Movement is optional.

Understanding is not.

Components Describe Behaviour

Components should communicate:

```
[text]
Hero Changed
```

Not:

```
[text]
Play Hero Animation
```

The Runtime Motion Resolver determines:

- timing,
- sequencing,
- curves,
- material response,
- accessibility.

Applications remain behaviourally simple.

Avoid Decorative Motion

Before implementing movement ask:

> *Would removing this reduce understanding?*

If the answer is:

No.

The movement probably does not belong.

Mosaic values meaningful movement over impressive movement.

Respect The Environment

Canvas should rarely move.

Atmosphere should evolve slowly.

Materials should respond naturally.

The environment should always feel calmer than the objects moving within it.

Runtime Owns Motion

Applications should never manually determine:

- easing,
- duration,
- sequencing,
- environmental response.

Applications communicate behaviour.

Runtime resolves motion.

This separation keeps the behavioural language consistent across the platform.

Plugins

Extensions should contribute:

- behaviour,
- information,
- artwork.

They should never contribute:

- transitions,
- easing,

- animation systems,
- Material Motion.

Every extension should inherit the Motion System automatically.

Platform Independence

Before implementing movement ask:

> *Would this still communicate the same behaviour on:*

- Web?
- Flutter?
- Television?
- Mobile?
- Future platforms?

Implementation may differ.

Behaviour should remain identical.

Performance

Performance optimisation should preserve behavioural meaning.

Preferred.

```
[text]
Same Behaviour

↓

Simpler Rendering
```

Avoid.

```
[text]
Different Behaviour

↓

Higher Performance
```

Optimise implementation.

Never behavioural communication.

Common Mistakes

Avoid the following.

Animation Thinking

Choosing animations before understanding behaviour.

Simultaneous Motion

Everything moving together.

Decorative Movement

Motion existing purely because it looks impressive.

Component Animation

Components independently defining transitions.

Platform Motion

Each client inventing different behavioural movement.

Atmosphere Animation

Environmental lighting changing without behavioural justification.

Motion Review Questions

Before implementing any movement ask:

- What behaviour changed?
- Does this preserve continuity?
- Does hierarchy remain clear?
- Do Materials respond naturally?
- Does Typography remain readable?
- Would this still feel like Mosaic?

If uncertainty remains...

Return to the behavioural model before writing animation code.

Motion Checklist

Every motion implementation should satisfy the following.

- [ ] Behaviour clearly initiated movement.
- [ ] Motion Hierarchy is respected.
- [ ] Material Motion remains coherent.
- [ ] Refraction follows Materials.
- [ ] Typography remains readable.

- [ ] Accessibility is preserved.
- [ ] Platform independence is maintained.
- [ ] Entertainment remains the centre of attention.

Final Guidance

The Motion System should eventually disappear from conscious thought.

Contributors should stop asking:

> *"How should this animate?"*

and instinctively begin asking:

> ***"How should the user's World evolve?"***

When every contributor naturally thinks in behavioural evolution rather than animation, the Motion System has achieved its purpose.

Users will simply feel that Mosaic responds exactly as they expected.

That effortless continuity is the defining characteristic of the Mosaic Motion System.

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`glossary.md`

Glossary

Purpose

This glossary defines the terminology introduced by **MDS-005 — Motion System**.

Unlike previous glossaries, these definitions describe the architectural language of behavioural motion rather than animation technology.

These definitions should remain stable regardless of:

- rendering engine,
- animation framework,
- platform,
- device.

Future specifications should reuse these terms consistently.

B

Behavioural Motion

The visible expression of behavioural change.

Behavioural Motion communicates:

- what changed,
- why it changed,
- what should happen next.

It exists because behaviour evolved.

Not because animation exists.

C

Continuity

The perception that the user's World remains persistent while behaviour evolves.

Continuity is one of the primary objectives of the Motion System.

Users should feel that experiences continue.

Not restart.

E

Environmental Motion

The quiet evolution of:

- Runtime Atmosphere,
- Canvas,
- environmental lighting,
- surrounding Materials.

Environmental Motion should remain almost imperceptible.

Users should feel it more than consciously observe it.

H

Hero Motion

The highest-priority movement within the Motion Hierarchy.

Hero Motion communicates major behavioural changes such as:

- Focus changes,
- Hero transitions,
- playback evolution.

Hero Motion should feel confident rather than dramatic.

M

Material Motion

The physical movement of Mosaic materials in response to behavioural change.

Material Motion communicates:

- depth,
- physicality,
- continuity.

It should never become decorative.

Motion Hierarchy

The ordered system through which behavioural importance determines movement priority.

The hierarchy consists of:

```
[text]
Hero

↓

Primary

↓

Supporting

↓

Peripheral

↓

Environmental
```

Movement follows this hierarchy.

Not interface structure.

Motion Resolver

The runtime subsystem responsible for transforming behavioural events into resolved movement.

Applications communicate behaviour.

The Motion Resolver determines implementation.

P

Platform Motion

The platform-specific implementation of the Mosaic Motion System.

Platform Motion preserves behavioural meaning while adapting to rendering technologies such as:

- Web
- Flutter
- SwiftUI
- Compose

Implementation changes.

Behaviour remains constant.

R

Refraction Motion

The temporal evolution of environmental light as it propagates through Mosaic materials.

Refraction Motion follows Material Motion.

It should never precede behavioural change.

Runtime Motion Resolution

The deterministic process that transforms behavioural events into platform-specific motion.

Runtime Motion Resolution evaluates:

- Behaviour,
- Hierarchy,
- Materials,
- Accessibility,
- Platform.

before producing resolved movement.

T

Temporal Continuity

The preservation of behavioural continuity across time.

Temporal Continuity ensures users perceive one evolving World rather than disconnected interface states.

Transition

A behavioural evolution between two meaningful states.

Within Mosaic, transitions communicate continuity rather than replacement.

U

Understanding

The successful communication of behavioural change.

Understanding is considered the primary success metric of the Motion System.

Motion exists to improve understanding.

Not visual richness.

Cross References

Specification	Primary Concepts
MDL-001 Vision	Companion, Immersion
MDL-002 Principles	Behaviour, Calmness
MDL-003 Mental Model	World, Focus
MDL-004 Interaction Model	Behavioural Evolution
MDL-005 Composition Model	Hierarchy
MDS-003 Material System	Material Motion, Refraction
MDS-004 Typography System	Editorial Continuity

Terminology Rules

Future contributors should:

- describe behaviour before animation,
- describe continuity before transitions,
- distinguish Motion from animation,
- distinguish Material Motion from Refraction Motion,
- avoid implementation-specific animation terminology within architectural documentation.

Motion terminology should remain independent from rendering technology.

Review Status

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`references.md`

References

Purpose

This document records the architectural influences and conceptual foundations that informed **MDS-005 — Motion System**.

Unlike implementation documentation, these references explain *why* Mosaic moves the way it does rather than prescribing animation techniques or platform APIs.

The Motion System intentionally combines ideas from:

- behavioural psychology,
- physical movement,
- environmental continuity,
- industrial motion,
- human perception,
- interaction design,

into one coherent behavioural language centred around understanding rather than animation.

Reading Order

Contributors should approach references in the following order.

1. MDL Specifications 2. Design Token Architecture 3. Colour System 4. Material System 5. Typography System 6. Motion System 7. Platform Implementations

The Mosaic Design Language remains the authoritative source.

External references provide inspiration rather than specification.

Internal References

MDL-001 — Vision

Provides:

- Companion philosophy
- Entertainment-first thinking
- Calm interaction
- Immersion

Motion should reinforce the feeling that the Companion understands what the user is trying to do.

It should never compete for attention.

MDL-002 — Principles

Provides:

- Behaviour Before Interface
- Content Leads
- Calm Interfaces
- Every Feature Earns Its Place

Motion is the direct implementation of these principles over time.

MDL-003 — Mental Model

Provides:

- World
- Focus
- Context
- Relationships

Motion exists because the World evolves.

It should never imply unrelated change.

MDL-004 — Interaction Model

Provides:

- Behaviour
- Continuity
- Focus transitions
- Context evolution

The Motion System is the physical expression of the Interaction Model.

Without MDL-004 there is no behavioural basis for movement.

MDL-005 — Composition Model

Provides:

- Hero
- Hierarchy
- Priority
- Anchors
- Density

Motion Hierarchy is intentionally derived from the Composition Hierarchy.

Movement reinforces understanding already established by Composition.

MDS-001 — Design Token Architecture

Provides:

- Runtime Tokens
- Resolution
- Semantic hierarchy

Runtime Motion Resolution extends the architectural resolution model established within MDS-001.

MDS-002 — Colour System

Provides:

- Runtime Atmosphere
- Colour Resolution
- Atmosphere Synthesis

Environmental Motion directly evolves Runtime Atmosphere.

Atmosphere should therefore never animate independently from behaviour.

MDS-003 — Material System

Provides:

- Acrylic
- Hero Material
- Refraction
- Light Transport
- Runtime Material Resolution

Material Motion and Refraction Motion are direct continuations of the Material System.

MDS-004 — Typography System

Provides:

- Editorial Hierarchy
- Reading Rhythm
- Runtime Typography

Motion should preserve editorial continuity.

Typography should remain readable throughout every behavioural transition.

Future Specifications

The following specifications directly depend upon MDS-005.

- MDS-006 Composition Engine
- MDS-007 Tile Framework
- MDS-008 Component Library

These specifications consume Motion.

They should never redefine behavioural sequencing independently.

Behavioural Design

The Motion System draws significant inspiration from behavioural design.

Primary influences include:

- causality
- continuity
- progressive disclosure
- object permanence
- predictable interaction

Movement exists to explain behaviour.

Not to entertain.

Human Perception

Several characteristics of perception influenced the Motion System.

Examples include:

- motion prediction
- spatial memory
- object permanence
- peripheral perception
- temporal continuity

The Motion Hierarchy is designed to work with these behaviours rather than against them.

Physical Motion

The Motion System intentionally borrows concepts from physical objects rather than digital animation.

Examples include:

- inertia
- settling
- diffusion
- environmental response
- momentum

The objective is perceived physicality rather than simulation accuracy.

Environmental Behaviour

One of the defining influences behind Mosaic Motion is environmental behaviour.

Objects move.

Light follows.

Atmosphere settles.

The environment remains coherent.

This relationship strongly differentiates Mosaic from conventional interface animation.

Accessibility

Accessibility is treated as a first-class architectural concern.

The Motion System intentionally assumes:

```
[text]
Behaviour
```

↓

```
Accessibility
```

↓

```
Motion
```

↓

```
Presentation
```

This ordering ensures movement never becomes a prerequisite for understanding.

Runtime Systems

Unlike traditional animation systems, Mosaic assumes behaviour changes continuously.

Examples include:

- Focus
- Context
- Playback

- Reading
- Runtime Atmosphere
- Material response

Motion therefore becomes a continuously evolving behavioural system rather than a collection of isolated transitions.

Platform Independence

The Motion System intentionally separates:

```
[text]
Behaviour

↓

Runtime Resolution

↓

Platform Motion

↓

Presentation
```

Every client should therefore communicate identical behavioural understanding while remaining free to implement movement using:

- CSS
- Flutter
- SwiftUI
- Compose
- future rendering technologies.

Mosaic-Specific Influences

The Motion System emerged directly from founder exploration.

Major architectural discoveries included:

- Motion should explain behaviour rather than animate interfaces.
- Environmental light should move independently from geometry.
- Materials should respond before the environment settles.
- Behaviour should always precede animation.
- Users should experience one continuous World rather than a sequence of screens.

Together these discoveries define the behavioural language unique to Mosaic.

Relationship To The Companion

Motion represents the physical body language of the Companion.

Conceptually.

```
[text]
Companion

↓

Behaviour

↓

Motion

↓

Understanding

↓

User
```

Words express intent.

Motion expresses confidence.

Together they create a Companion that feels calm, deliberate and predictable.

Normative References

Required reading before contributing to MDS-005.

- MDL-001 Vision
- MDL-002 Principles
- MDL-003 Mental Model
- MDL-004 Interaction Model
- MDL-005 Composition Model
- MDS-001 Design Token Architecture
- MDS-002 Colour System
- MDS-003 Material System
- MDS-004 Typography System

Together these specifications define the conceptual foundation of the Motion System.

Informative References

Future contributors may also wish to review:

- MDS-006 Composition Engine
- MDS-007 Tile Framework

- MDS-008 Component Library

These specifications describe how behavioural motion becomes runtime interaction throughout the Mosaic platform.

Living Document

This reference list should remain intentionally concise.

References should only be introduced when they materially influence:

- behavioural architecture,
- temporal continuity,
- runtime motion,
- implementation boundaries.

The objective is to preserve architectural reasoning rather than catalogue animation literature.

Completion

This concludes **MDS-005 — Motion System**.

The next specification in the Mosaic Design System is:

> ***MDS-006 — Composition Engine***

Where MDS-005 defines **how the world moves**, MDS-006 defines **how the world is assembled at runtime**.

It formalises:

- Runtime Composition Engine
- Composition Solver implementation
- Expression resolution
- Adaptive layouts
- Runtime hierarchy
- Behaviour orchestration
- Graph-driven composition
- Multi-device rendering
- Expression pipelines

MDS-006 is expected to become the architectural heart of Mosaic, where every concept introduced by the MDL and earlier MDS specifications converges into one runtime system capable of constructing the user's World dynamically.